

1. As already discussed in the class, create data splits in your data. Use random sampling to choose sentences and create the validation and test sets.
 - a. Validation Set - 1000 sentences
 - b. Test Set - 1000 sentences
 - c. Training Set - Remaining Sentences
2. By now, you have already built the following models,
 - a. Unigram Model
 - b. Bigram Model
 - c. Trigram Model
 - d. Quadrigram Model

Implement Good Turing Smoothing for all the models.

For unseen n-grams, the cumulative probability to be distributed to all unseen n-grams = N_1/N , where N = total number of seen n-grams and N_1 =number of times a n-gram has occurred only once

Good Turing smoothing for individual unseen n-gram $P_{\text{unseen}} = (N_1/N)/(\text{number of unseen n-grams}) = (N_1/N) / (V^n - N) = N_1 / N * (V^n - N)$ where V = vocabulary size, where $n \geq 2$ for unigram model it can be = V - unique seen unigrams (U), $P_{\text{unseen}} = N_1 / N * (V - U)$

Compute the probability of each sentence in the validation and test sets using the smoothed models.

3. Show a table with top 100 frequencies as the following:

C (MLE)	N_c	C^*
0		
1		
...		

4. Implement deleted interpolated smoothing technique for the quadrigram model and find the best parameters.